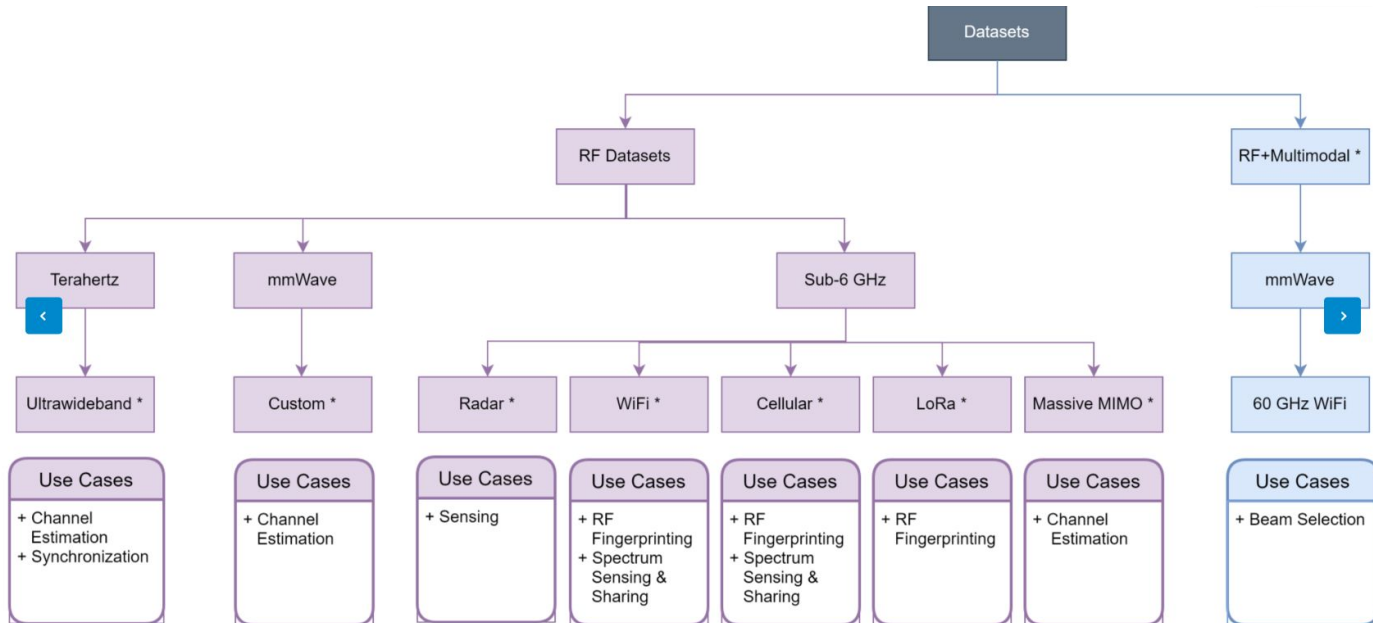




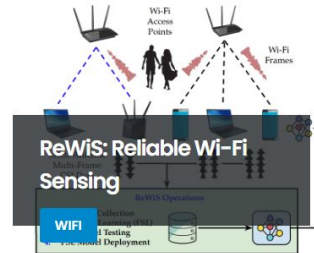
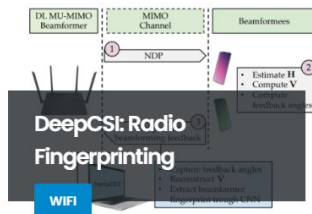
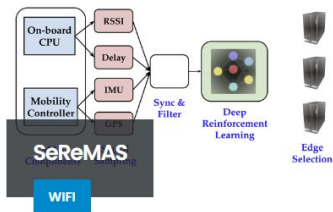
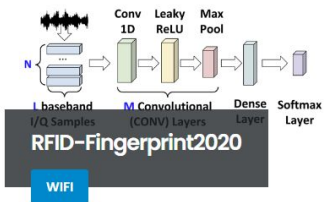
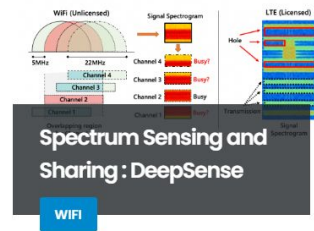
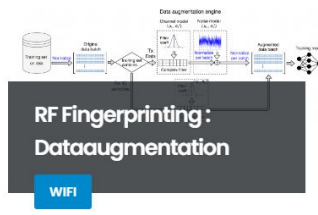
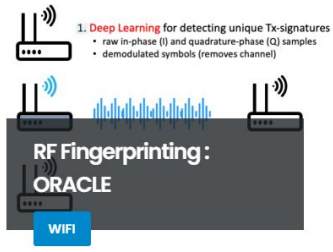
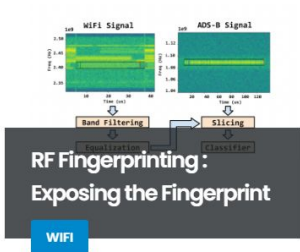
RF Data Factory Experimental Setups

Aadi - Bethlehem Central High School

Dataset Organization

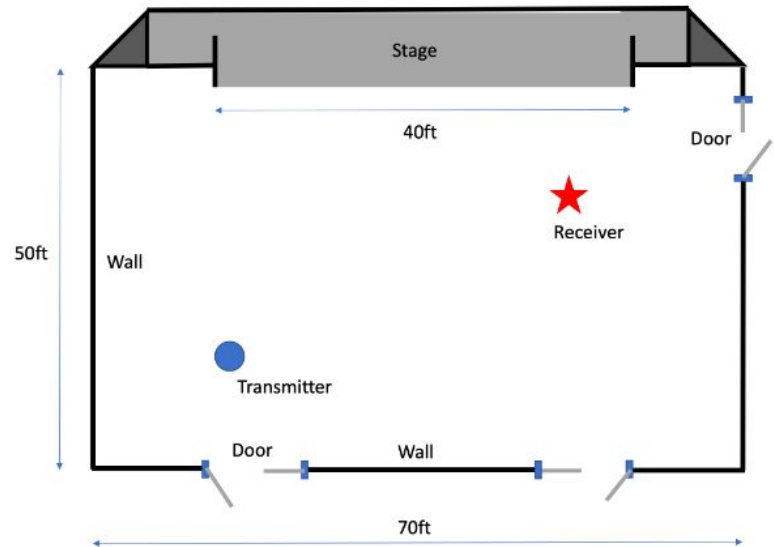


Wifi Datasets



RF Fingerprinting of Bit-similar USRP X310 Radios

- Detects a unique radio from a large pool of bit-similar devices (hardware, protocol, physical address, MAC ID) using IQ samples at the physical layer
- Setup:
 - Transmitters: USRP X310 radios which emit IEEE 802.11a frames generated from MATLAB WLAN Systems box
 - Receiver: Fixed USRP B210
 - Payload: random but all have the same address field - streamed to selected SDR for over-the-air wireless transmission
 - Receiver SDR receives samples at 5MS/s at a frequency of 2.45 GHz
 - 20 million samples per each radio
 - Collecting raw I/Q data



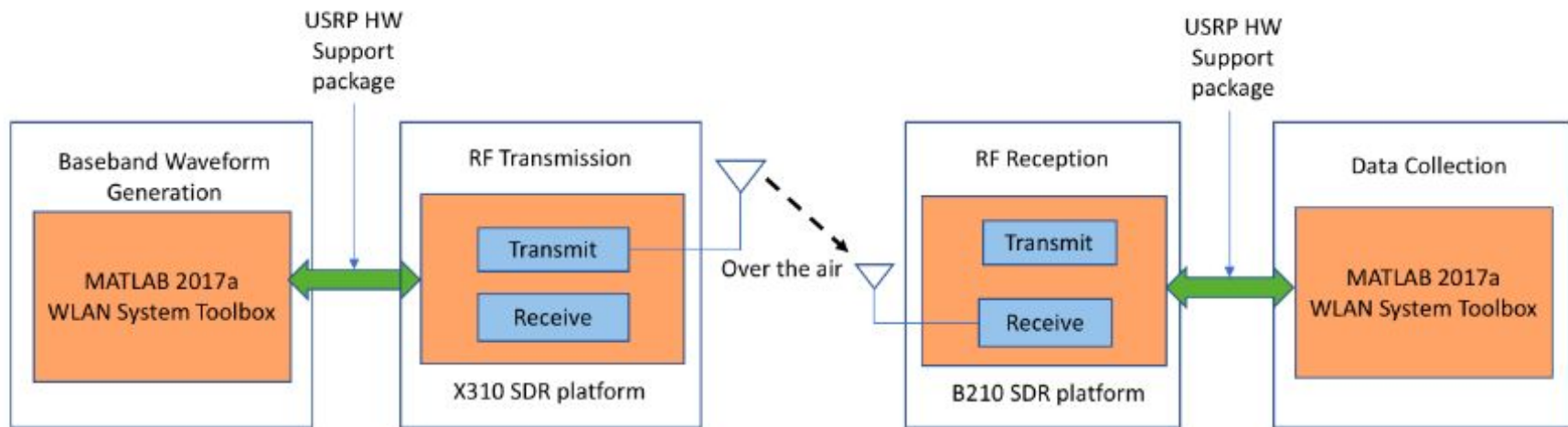


Fig. 3: Experimental setup for data collection using SDR

Cellular

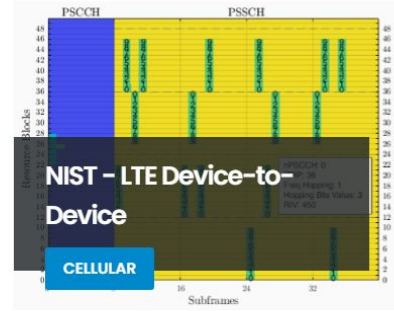


RF Fingerprinting: POWDER PAWR

CELLULAR

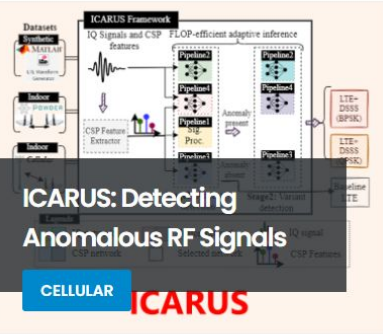
Spectrum Sensing and Sharing : O-RAN for Data-driven NextG Cellular Networks

CELLULAR



NIST - LTE Device-to-Device

CELLULAR



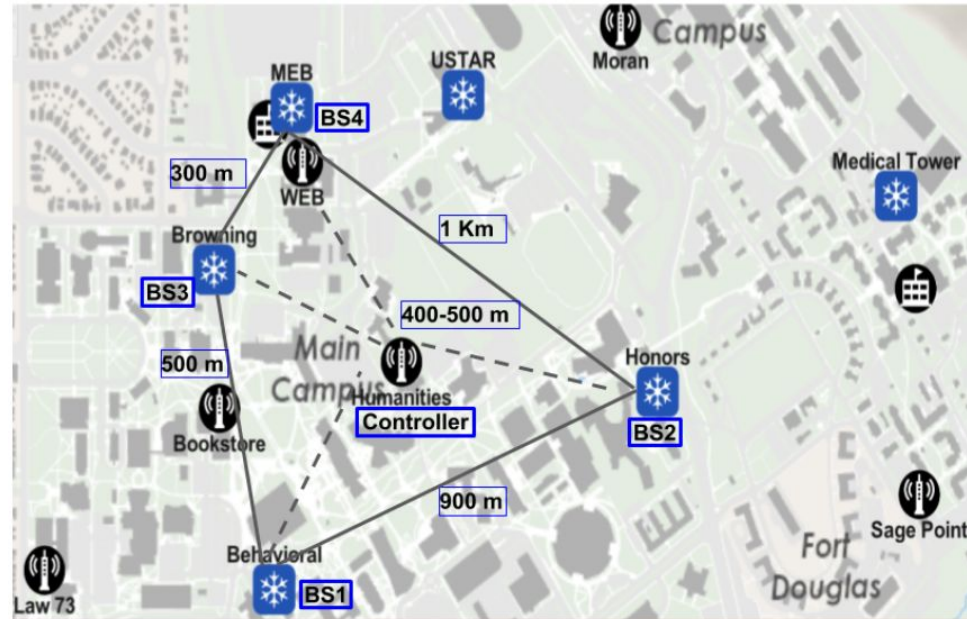
ICARUS: Detecting Anomalous RF Signals

CELLULAR

ICARUS

POWDER RF Fingerprinting

- Collect IQ samples from 4 different buildings with a fixed endpoint
- Transmit IEEE 802.11a, LTE, or 5G-NR frames generated via MATLAB
- Random payload with same address field
- Receiver SDR (USRP B210) works at 2.685 GHz
- Sample incoming signals at 5MS/s for Wifi and 7.69MS/s for LTE and 5G
- 5 sets of 2s IQ samples for each link and waveform
- Collected independently over 2 days



LORA

LoRa Radio Dataset

- Grid of 8x8 VERT2450 antennas mounted to the ceiling of a 2240 sq ft indoor office space
- Each antenna is cabled to a programmable SDR through a low-attenuation coax cable
 - Enables sub 6GHz 5G and beyond research
- 24 SDRs controlled by 12 Dell Power Edge R340 running Ubuntu
- Similar setup was done outside
 - Receiver and gateway kept indoor
- Data: Raw IQ samples
- Carried out over several days

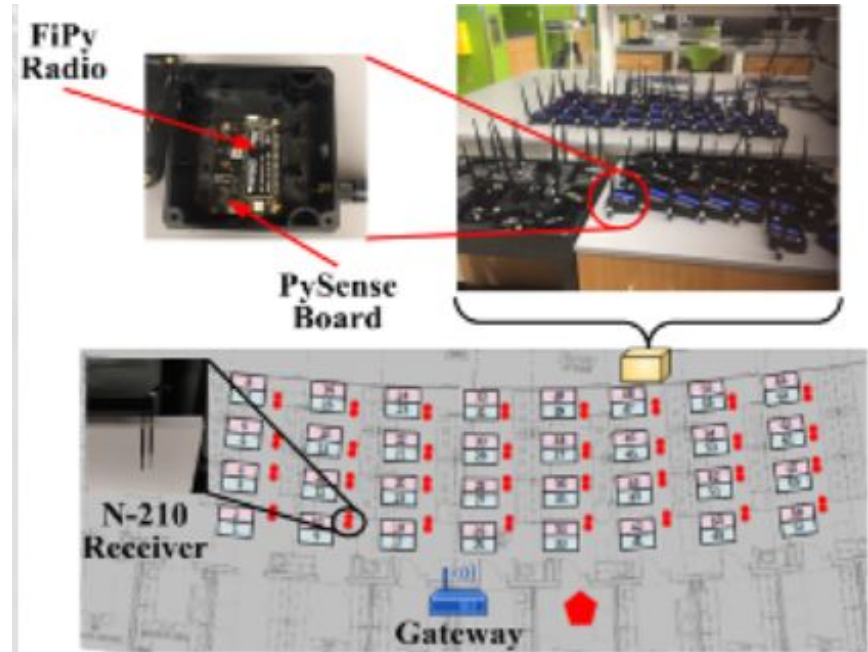


Figure 1



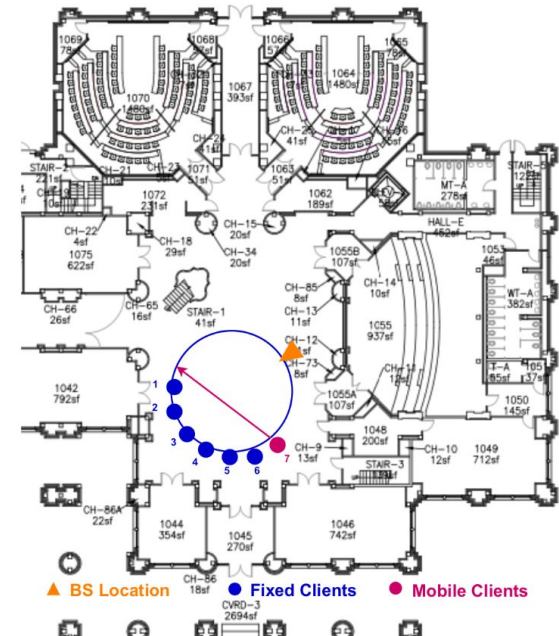
Data Collection Continued

- “The radios in our setup operate on a carrier of 902.3 MHz in the 915 MHz ISM band. Each transmits ten consecutive bursts of packets in each location. Each burst is separated by 1 second. The burst consists of 100 consecutive packets separated by 10 ms. Each packet contains the payload information carrying the temperature, the humidity, and the device voltage readings. A USRP N-210 or B-210 is synchronized to receive the transmitted packets in the data collection testbed.”

Massive MIMO

Indoor Mobility Channel Measurement for Massive MIMO

- 64-antenna Massive MIMO base station with 7 software defined clients spread across a large room
- 6 clients were placed in a circle 15 meters away from the base station
- Remaining client was placed on a robot
- Robot moved from the first client to the last client in a straight path at various speeds - 0.5m/s, 1.0m/s, and 2.0m/s
- Mobile node always facing towards the base station
- Time orthogonal uplink pilots from client to base station
 - 802.11 LTS OFDM signal
 - 52 non-zero subcarriers





Fixed UE

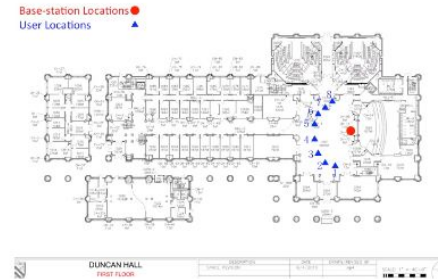
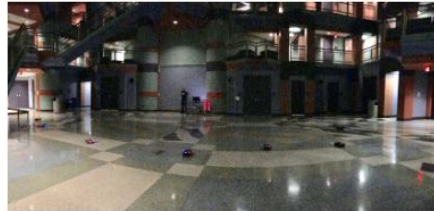
UE Trajectory

Mobile UE

Argos Channel Survey

- Many antenna MU-MIMO channel measurement
- Channel traces in UHF, 2.4GHz, and 5GHz bands

- Up to 104 base-stations serving 8 users
- Varying mobility (i.e pedestrians)
- Measured channel stability across users
- [More info](#)



Multimodal

Fast mmWave

- Goal: to use cameras and images to guide a receiver and transmitter to find the best beam configuration
 - Beam sweeping takes precious time
- Transmitter and receiver mounted on horizontal sliders
- 2 GoPros are used to monitor the room
- Obstacle blocking the LOS path between Tx and Rx (33 cm × 88 cm × 3 cm and 33 cm × 88 cm × 10 cm)
 - Causes 30 dB and 4 dB attenuation
- Record link quality of different beams and two pictures

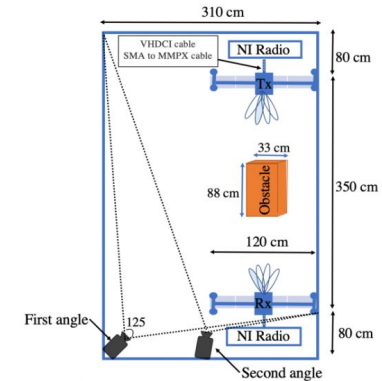
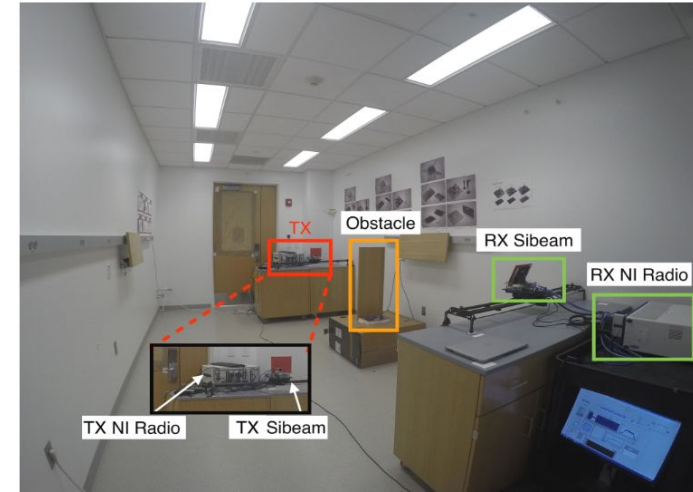


Fig. 3: A diagram representing experiment parameters

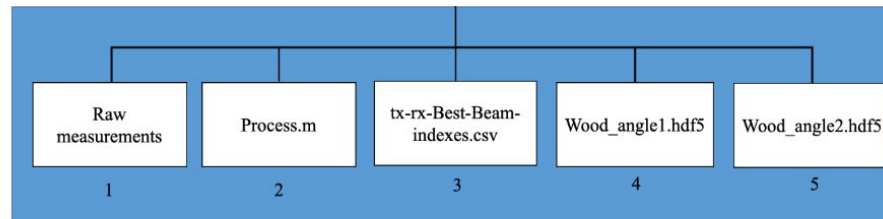


Fig. 4: The list of files included in the dataset

1. **Raw measurements:** It is a folder containing raw measurements of link qualities. For each beam pair, we measure 50 samples of received signal SNR. The transmitter and receiver have a codebook of 13 elements each, leading to 169 different beam pair configuration possibilities. The first column shows the measured received SNR and the second column in each CSV file shows the beam configuration at Tx and Rx. While moving from one configuration to another one, we write zeros to the CSV files.
2. **Process.m:** We use equation (3) to define the best beam configuration. This Matlab code gets the raw measurements as input and generates the best beam configuration indexes at the transmitter and receiver.
3. **Tx-rx-Best-Beam-indexes.csv:** Each location on the slider is defined by a case id (see section III). The case ids and associated best beam pair (the labels) are defined in this file. The labels are also included in (4) and (5).
4. **Cardbox_angle1.hdf5:** This hdf5 file is the main file, including the dataset itself. Fig. 5 shows the hierarchical structure used to generate hdf5 files.



FLASH: Federated Learning for Automated Selection of High-band mmWave Sectors

- Fast sector-steering in mmWave vehicle scenarios are challenging
 - standard-defined exhaustive search cannot be completed in short contact times
- Machine learning using LIDAR, GPS, and camera images
- Test bed:
 - Autonomous vehicle with GPS and LIDAR connected to an onboard computer via low latency ethernet - have their own IP addresses
 - Car acts as a receiver
 - Stationary transmitter on the side of the road
 - GoPro cameras also onboard car and are connected via wifi and USB
 - Custom python scripts are used to control camera recordings and labeling

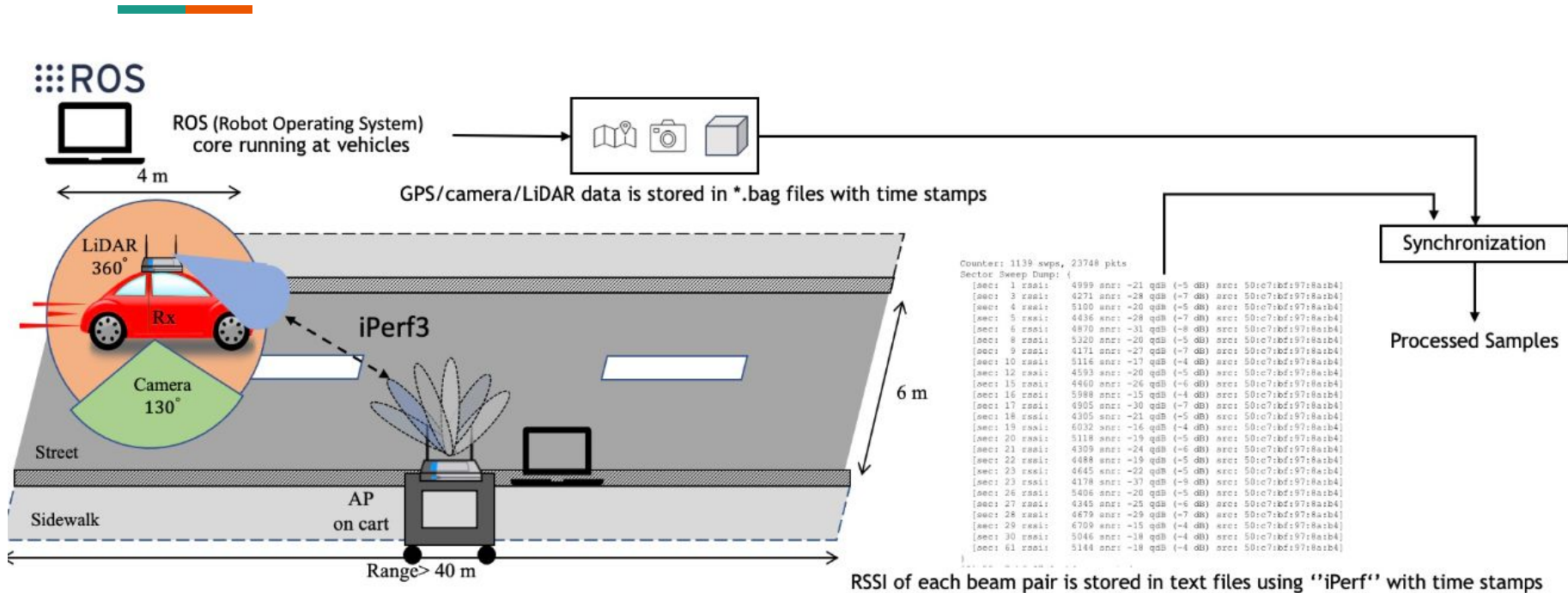


Fig 4: FLASH testbed

- 
- **mmWave radio:** We use TP-Link Talon AD7200 tri-band routers with Qualcomm QCA9500 IEEE 802.11ad Wi-Fi chips as both the BS and Rx at the 60 GHz frequency. The default codebook includes sector IDs from 1 to 31 and 61-63 for a total of 34 sectors; the sectors with IDs of 32 to 60 are undefined. We gain access to PHY-layer characteristics of AP and RX using the open-source Linux Embedded Development Environment (LEDE) and Nexmon firmware patching released by [Steinmetzer et al.](#) We record the time-synchronized RF ground truth data as data transmission rate and received signal strength indication (RSSI) at each sector.
 - **GPS:** The MyLincoln Touch On-board GPS provides coordinate readings at sampling rate of 1Hz.
 - **Camera:** Camera sensor (GoPro Hero4) with a field-of-view (FOV) of 130 degrees provides video readings at sampling rate of 30fps.
 - **LiDAR:** Velodyne (VLP 16) LiDAR with a FOV of 360 degrees records point clouds at sampling rate of 10Hz.



Talon AD7200
Multi-Band Wi-Fi
Router



MyLincoln
Touch On-board
GPS



GoPro Hero 4



Velodyne LiDAR